

ANALYSTLAB AFRICA — HealthConnect Clinic Experience Lab

HealthConnect Analytics Report

Data Analytics Track — Week 5: Exploratory Analysis, KPI Development & Business Insights

Executive Summary

This report presents the Week 5 analysis of HealthConnect Clinic's appointment data, examining patterns behind patient no-shows and translating them into measurable KPIs and actionable business recommendations. The overall no-show rate across the dataset is 48.46%, meaning close to half of all scheduled appointments are missed. The strongest driver identified is booking lead time: no-shows rise from 29.01% for appointments booked 0–9 days in advance to 67.94% for appointments booked 50–60 days out. Reminders show a modest protective effect (47.36% no-show rate with a reminder vs 51.39% without), and demographic and distance-related factors show weaker but still notable patterns. Five business recommendations and six limitations are set out below to guide HealthConnect's next steps.

Key Findings

- Overall no-show rate: 48.46% of all scheduled appointments — a high baseline that justifies dedicated intervention.
- Booking lead time is the strongest driver found: no-show rate climbs steadily from 29.01% (0–9 days) to 67.94% (50–60 days).
- Reminders are associated with a lower no-show rate (47.36% with a reminder vs 51.39% without) — a modest but consistent effect.
- Demographic differences (age, gender) are relatively moderate, though some groups recorded higher no-show rates than others.
- Distance to the clinic may contribute to no-show behaviour, but missing values and inconsistent grouping limit how strong a conclusion can be drawn at this stage.

Business Recommendations

Based on the analysis of appointment attendance and no-show patterns, the following recommendations are proposed for HealthConnect:

1. Strengthen reminder systems

HealthConnect should strengthen its appointment reminder process, as appointments with reminders had a lower no-show rate (47.36%) compared with appointments without reminders (51.39%). The clinic could use automated SMS or other appropriate reminder channels, particularly for appointments identified as higher risk.

2. Introduce additional follow-up for appointments booked far in advance

The analysis found a substantial increase in no-show rates as booking lead time increased. No-shows increased from 29.01% for appointments booked 0–9 days in advance to 67.94% for appointments booked 50–60 days in advance. HealthConnect should consider sending additional reminders or confirmation requests for appointments scheduled several weeks in advance.

3. Develop targeted attendance interventions

Although demographic differences were relatively moderate, some groups recorded higher no-show rates than others. HealthConnect could monitor demographic patterns and use targeted communication strategies where appropriate rather than applying the same intervention to all patients.

4. Monitor no-show performance regularly

HealthConnect should establish regular monitoring of the 48.46% overall no-show rate and track changes over time. A Power BI dashboard can help management monitor appointment outcomes, identify emerging patterns, and evaluate whether interventions are improving attendance.

5. Investigate distance-related barriers further

Distance may contribute to appointment attendance behaviour, but the current analysis does not provide sufficiently standardized distance groups for strong conclusions. HealthConnect should collect and analyse distance using consistent distance bands and consider whether transportation difficulties or accessibility issues contribute to missed appointments.

Limitations

1. Observational analysis

The analysis identifies patterns and associations within the appointment data but does not establish that one factor directly causes a patient to miss an appointment.

2. Distance data limitations

Distance to the clinic contained missing values and was not easily grouped into standardized distance bands. This limits the reliability of conclusions regarding the relationship between distance and no-show behaviour.

3. Unequal group sizes

Some categories may contain substantially different numbers of appointments. A percentage can therefore be misleading when a group has a very small number of observations.

4. Limited demographic variables

The dataset contains variables such as age and gender, but other factors that could influence attendance may not be captured, such as transportation availability, employment schedules, socioeconomic circumstances, or reasons for cancelling.

5. Reminder association does not prove effectiveness

Although appointments with reminders had a lower no-show rate, the analysis does not prove that the reminders themselves caused the reduction. Other differences between patients who received and did not receive reminders may have influenced the results.

6. Historical dataset

The findings reflect the appointments contained in the analysed dataset and may not represent future appointment behaviour. HealthConnect should continue monitoring its data to determine whether these patterns remain consistent over time.